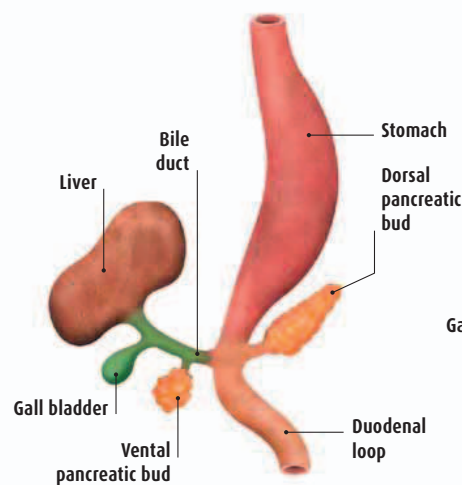


Q Is it true that my baby's entire digestive system begins as a simple tube?

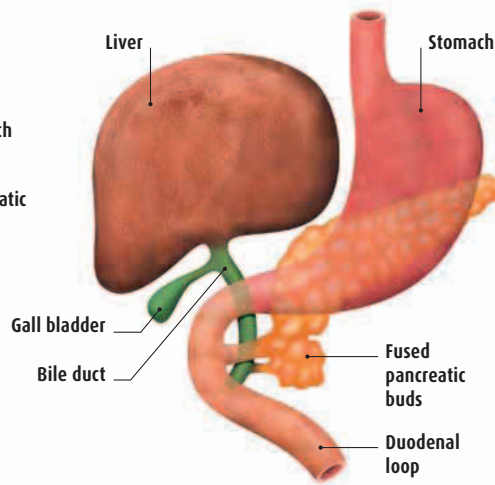
The whole digestive system forms its structure by 12 weeks. By the time he is born, your tiny new baby will house an incredible 8 ft (2.5 m) of intestines.

Your baby's digestive system begins with a "gut tube" that forms from the endoderm layer of cells (the lower layer) and connects an opening at the top of the tube (which will connect to your baby's mouth) to an opening at the bottom (eventually your baby's anus). The gut tube then develops three sections—the foregut, midgut, and hindgut—and each of these forms specific organ buds. Between six and eight weeks pregnant, the foregut develops a bulge that will become your baby's stomach. All three sections of the gut tube continually lengthen so that by the time you are nine weeks pregnant, there's so much

bowel tissue (the bowel is the collective name for the small and large intestines) that there isn't room for it to fit inside the tiny embryo. Instead, it pushes out of a hole in your baby's abdomen so that it occupies some of the space in the umbilical cord. By 10 weeks pregnant, the baby's stomach will begin to produce digestive juices—a milestone in his transition from embryo to fetus. By the time you're 12 weeks pregnant, your baby's body is large enough for his bowel to come back inside his body and his umbilical cord is empty again, aside from the vessels that belong there.



1 At nine weeks The main tube of the gut develops specialized structures that branch off from it. The early pancreas is made up of two separate buds.



2 At 10 weeks The pancreatic buds have fused. The liver and stomach are much larger, and the duodenum (part of the small intestine) is lengthening.

Q Does my baby have functioning kidneys? If so, where does the urine go?

By 13 weeks of pregnancy, your baby's urine can pass out of his newly forming kidneys, through his developing ureters, and into his tiny bladder. This urine is released into the amniotic fluid. Your baby learns to swallow at around the same time, so drinks the amniotic

fluid, swallows his urine, fills his bladder again, and so the process goes on. Although this might sound unappetizing, fetal urine is very diluted and most waste has already been filtered out of your baby's system via the placenta, which is then expelled by your waste system. The kidneys develop from 10 weeks onward, multiplying thousands of filtration cells. The ureters that collect the urine branch into the kidneys, a process that takes until 32 weeks, when there are around two million branchings.

Q Does my baby's bowel work in the uterus?

As soon as your baby's bowel is fully formed the process of making meconium—his first poop—begins. Meconium consists of waste products such as cells from the bowel lining, amniotic fluid, mucus, and bile. It builds up and is slowly forced into the large intestine—a process helped by the peristaltic motion of your baby's bowel from 20 weeks. At roughly the same time, the muscle around his anus—the anal sphincter—tightens up, keeping the meconium in so that he doesn't poop inside the uterus.

0 Babies have no **kneecaps** when they are born. These will develop in infancy.

1 Only one organ, the **auditory ossicles** in the ear, is full size at birth.

33% A new baby's **kidneys** are just a third of full adult size.

270 Babies have more **bones** than adults, who have 206. Many bones fuse as they grow.

30,000 The number of **taste buds** a baby is born with reduces to 10,000 by adulthood.